

Data Storage and State Management

MongoDB, Redis, MinIO, Weaviate, checkpoints, and browser state.

Eurskem AI · Agentic Workflow Platform

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Derived from the live repository

Source of truth. Inventories and symbol tables are generated from the current checkout. Narrative explains observed structures and does not replace executable contracts.

Storage responsibility matrix

Store	Data	Role
MongoDB	Users, workflows, conversations, runs, histories, policies, costs	Durable document source of truth
Redis	Rate limits, leases, SSE, pub/sub, checkpoints	Distributed coordination and replay
MinIO	Uploads, workflow files, generated artifacts	Protected binary/object storage
Weaviate	Chunks, embeddings, metadata, hybrid retrieval	Search index constrained by owner scope
Browser memory	Current mode, loaded data, forms	Ephemeral presentation state
local/session storage	Preferences and attachment handoff	Non-authoritative client convenience

File	Responsibility
<code>app/db/migrations.py</code>	Versioned, idempotent MongoDB data migrations.
<code>app/db/mongo.py</code>	Async MongoDB client and manifest collection.
<code>app/retrieval/strategies.py</code>	Replaceable dense, sparse and hybrid Weaviate retrieval strategies.
<code>app/retrieval/weaviate_client.py</code>	Weaviate client and schema definition.
<code>app/storage/__init__.py</code>	Python module.
<code>app/storage/minio_client.py</code>	S3-compatible object storage client.
<code>app/workflow/builder_store.py</code>	Durable, file-backed state used by the Workflow Builder.
<code>app/workflow/chat_conversation_store.py</code>	Durable, owner-scoped Business Chat conversations and transcript messages.
<code>app/workflow/chat_workflow_store.py</code>	Owner-scoped persistence for private Business Chat workflows.
<code>app/workflow/claim_verifications.py</code>	Durable storage for claim-verification results (see <code>app.evidence.claim_verification</code>).
<code>app/workflow/file_inputs.py</code>	Workflow file-input contracts, validation, and object-storage helpers.
<code>app/workflow/prompt_template_store.py</code>	Owner-scoped custom prompt templates plus immutable built-in templates.
<code>app/workflow/run_chat_store.py</code>	Durable storage for "Ask AI about this run" conversations.
<code>ui/src/hooks/useRunSocket.ts</code>	Reset state because this effect owns a new external WebSocket lifecycle.
<code>ui/src/modes/knowledge/CollectionContext.tsx</code>	Applies a freshly fetched list, keeping the stored choice when it still
<code>ui/src/modes/knowledge/collectionStore.ts</code>	The one selected Collection, shared by every Knowledge Studio tab.
<code>ui/src/modes/studio/Cockpit.tsx</code>	Keep launch-state fields grouped above: they all come from one stable
<code>ui/src/modes/studio/builder/binding.ts</code>	What a config field's raw stored value actually means to a human.
<code>ui/src/modes/studio/business-chat/chatWorkspaceStorage.test.ts</code>	React/TypeScript module.

File	Responsibility
<code>ui/src/modes/studio/business-chat/chatWorkspaceStorage.ts</code>	Browser-local notes are optional; storage failures must not block Chat.
<code>ui/src/modes/studio/cockpit-state.test.ts</code>	join is still reachable via `left`, so it must NOT be skipped even
<code>ui/src/modes/studio/cockpit-state.ts</code>	'pending' is the internal value for the user-facing "Waiting" status —
<code>ui/src/modes/studio/cockpit/useResizablePanel.ts</code>	Drag-to-resize a panel's width, persisted per storageKey so it survives reloads/navigation.
<code>ui/src/modes/studio/file-artifact.ts</code>	Detect a downloadable object-store key inside a node's output.
<code>ui/src/modes/studio/library/localState.test.ts</code>	This project's vitest runs in the default 'node' test environment (no
<code>ui/src/modes/studio/library/localState.ts</code>	Favorites and "recently used"/"continue where you left off" are stored in
<code>ui/src/modes/studio/run-history/run-list-utils.ts</code>	Display label for a RunStatus — distinct from cockpit-state's NodeStatus labels.